

EduGuardian AI 2.0

Comprehensive System Architecture, AI Algorithms & Operational Guide

Version: 2.0-Production | Institution: NMAM Institute of Technology (Nitte) | Date: September 01, 2026

1. Executive Summary & Core Mission

EduGuardian AI 2.0 is an enterprise-grade academic resilience and early intervention platform designed for higher education institutions. Traditional educational portals are passive repositories of examination marks and attendance logs. EduGuardian transforms raw institutional records into a **proactive, predictive, and multi-agent AI safety net** that detects academic vulnerability weeks before formal examinations.

The platform balances two foundational principles: **Deep Diagnostic Precision for Faculty** and **Psychological Safety for Students**. While educators gain access to explainable SHAP-grounded risk scores and intervention pipelines, students are protected from demoralizing risk labels, receiving instead personalized study plans, constructive guidance, and AI mentoring.

2. End-to-End System Architecture

EduGuardian is built as an orchestrated multi-service architecture running in fully isolated Docker containers. The stack coordinates authentication, live university portal adapters, asynchronous AI agent graphs, and specialized React dashboards.

Service	Port	Technology Stack	Core Functionality
db	:5432	PostgreSQL 15 Alpine	Multi-database engine (eduguardian & eduguardian_chatbot)
edu-backend	:5000	FastAPI, SQLAlchemy, PyJWT	Main REST API, portal scraping adapter, risk calculation engine, faculty quotas
gateway	:8000	FastAPI, LangGraph, AsyncPG	AI Chatbot orchestrator, multi-turn state graph, three-stage safety guardrails
agent-insight	:8001	FastAPI, official a2a-sdk	A2A Microservice: Computes subject-level strengths, root weaknesses, and habits
agent-planner	:8002	FastAPI, official a2a-sdk	A2A Microservice: Generates structured, day-by-day adaptive study schedules
agent-coach	:8003	FastAPI, official a2a-sdk	A2A Microservice: Empathetic recovery tutor providing Socratic step-by-step guidance
student-ui	:3001	React, Vite, Nginx	Student Portal: Marks cards, subject attendance, assignments, and guidance
admin-ui	:3002	React, Vite, Nginx	Teacher & Admin Portal: Live risk roster, mentor assignment, resource dispatch
chatbot-ui	:3000	React, Vite, Nginx SSE Proxy	AI Chat Interface: Real-time streaming conversational coaching and quiz sessions

3. Real-Time University Portal Data Integration

Unlike systems relying on static mock data, EduGuardian incorporates an asynchronous **Portal Adapter** (`portal_adapter.py`) that connects directly to the institutional **University Solutions Portal** over HTTPS.

Authentication & Data Extraction Pipeline:

- **Secure Handshake:** The student provides their mobile number and portal credentials. The backend initiates an HTTPS session, obtains session cookies, and handles portal redirects.
- **Deep HTML Scraper & Parser:** Extracts student enrollment (USN, Degree, Department, Semester, Section), verified subject registrations, continuous internal evaluation (IA) scores, real-time subject-wise attendance logs, and completed semester marks cards (Semesters 1 through 6).
- **Summer / Supplementary Examination Resolver:** Discovers historical supplementary exam sessions (e.g. JUNE2024 (Summer)), maps cleared subject codes back to their root semester, updates the effective SGPA, and tracks cleared failure counts.
- **Server-Side Caching & Zero-Password Persistence:** Extracted StudentContext is cached in PostgreSQL (`portal_student_contexts`) with 1-hour TTL. Passwords are never saved, logged, or exposed.

4. Predictive Academic Risk Engine (Multivariate Formulation)

The risk engine (academic_risk_engine.py) evaluates academic vulnerability using a multi-horizon synthesis that combines long-term historical marksheet resilience with short-term current-semester indicators.

Multi-Horizon Risk Synthesis Equation:

$$R_fused = (w_hist \times S_foundation) + (w_live \times S_live_velocity)$$

- When only attendance is available: **w_hist = 0.70, w_live = 0.30** (prevents high attendance from masking prior failures).
- When full signals (attendance, quizzes, assignments, LMS) exist: **w_hist = 0.45, w_live = 0.55**.

Quantitative Weighting Matrix & Indicators:

Category	Signal / Metric	Weight / Trigger	Risk Impact & Rule Description
Historical Foundation	Cumulative CGPA	Baseline (0-100)	≥8.5: 8 pts 7.5-8.49: 16 pts 6.5-7.49: 36 pts 5.5-6.49: 56 pts <5.5: 78 pts
Historical Foundation	SGPA Velocity Dip	+30.0 / +18.0 pts	Latest SGPA <5.0 triggers +30 pts (Critical Dip); <6.0 triggers +18 pts; Declining trend: +18 pts
Historical Foundation	Active Backlogs	+22.0 pts / arrear	≥2 active backlogs escalates student directly to HIGH RISK
Historical Foundation	Cleared Supplementary Arrears	+8.0 pts / cleared	Prior failures indicate conceptual prerequisites gap; ≥2 cleared backlogs triggers Medium/High
Historical Foundation	Borderline Grade Density	+10.0 pts	≥4 subject grades with D, E, P, or GP ≤ 5.0 indicates marginal comprehension across semesters
Live Semester Signals	Real-Time Attendance %	40% live weight	<65%: 94 pts (Critical Shortage); 65-74%: 76 pts (Sub-75% Cutoff); 75-84%: 40 pts; ≥85%: 10 pts
Live Semester Signals	Internal Quiz Average	30% live weight	<45%: 88 pts (Sub-passing internal score); 45-64%: 50 pts; ≥65%: 10 pts
Live Semester Signals	Missed Assignments	15% live weight	≥2 missed: 85 pts; 1 missed: 45 pts; 0 missed: 10 pts
Live Semester Signals	LMS Digital Activity	15% live weight	<40% portal activity: 75 pts; 40-69%: 40 pts; ≥70%: 10 pts

Three-Tier Risk Classification & Calibrated Recovery Probability:

- **HIGH RISK (R_fused ≥ 50.0 OR Latest SGPA < 5.0 OR Backlogs ≥ 2 OR Att < 65%):** Immediate faculty mentor assignment and structured intervention mandatory. Recovery probability: **30% – 60%**.
- **MEDIUM RISK (R_fused 26.0–49.9 OR Cleared Arrears ≥ 1 OR CGPA < 6.8 OR Att 65–74%):** Targeted study plan and milestone check-in advised. Recovery probability: **62% – 80%**.
- **LOW RISK (R_fused < 26.0, CGPA ≥ 7.0, SGPA ≥ 6.5, Att ≥ 75%):** On track for academic distinction. Recovery probability: **84% – 96%**.

5. Role-Based Risk Isolation & Psychological Protection

A central design requirement of EduGuardian is the **strict segregation of risk metrics between faculty and student views**. Educational research demonstrates that displaying negative classifications (such as 'At Risk' or 'Declining Trajectory') creates student anxiety and self-fulfilling negative performance loops.

Portal / Perspective	What is Displayed (Allowed)	What is Strictly Masked (Forbidden)
----------------------	-----------------------------	-------------------------------------

Teacher & Admin Portal (:3002)	• Full 3-Tier Risk Badges (High, Medium, Low) • Exact Risk Score (e.g. 68.8 / 100) & Confidence • SHAP Feature Attribution & Root Factors • Attendance Shortage Warnings (<75% / <65%) • Mentor Assignment & Resource Upload Controls	<i>None. Faculty requires complete, transparent diagnostic data to make timely pedagogical interventions.</i>
Student Portal (:3001)	• Cumulative CGPA and Semester SGPAs • Verified Subject-wise Attendance Logs • Active Enrolled Term (e.g. Semester 5 ISE) • Total Credits Earned (e.g. 96 Credits) • Constructive Guidance & Mentoring Actions	• NO Risk Scores or Risk Levels • NO 'Declining' Trajectory labels in red • NO SHAP mathematical breakdown • NO Demoralizing or punitive banners

6. Multi-Agent Chatbot & A2A Microservices Architecture

EduGuardian's conversational intelligence is orchestrated by a **LangGraph StateGraph** in the FastAPI Gateway, delegating domain-specific tasks to specialized microservices over Google's official **Agent-to-Agent (A2A) SDK**.

The Three Autonomous A2A Agents:

1. **Student Insight Agent (agent-insight:8001)**: Computes historical marksheet mastery, analyzes subject-level low grade patterns, and identifies root academic struggle areas.
2. **Adaptive Study Planner Agent (agent-planner:8002)**: Synthesizes exam schedules and syllabus priorities to construct a milestone-driven, multi-week study timetable with realistic daily time budgets.
3. **Recovery Coach Agent (agent-coach:8003)**: Acts as an empathetic Socratic tutor. It breaks down complex technical questions (e.g. Dijkstra algorithm, OS paging, ML cost functions) into bite-sized questions and evaluates student comprehension.

7. Three-Stage AI Safety & Grounding Guardrails

To ensure safe and reliable pedagogical interactions, every student query flows through a deterministic three-stage guardrail pipeline before and after LLM execution.

Guardrail Stage	Execution Point	Mechanism & Protections Applied
Stage 1: Input Guardrail	Pre-Orchestration (Immediate On Request)	• Jailbreak Defense: Detects prompt injection, system prompt leakage, and role-override attempts. • Off-Topic Deflection: Refuses non-educational inquiries (entertainment, coding exploits). • Mental Health Support: Flags severe emotional distress and directs student to institutional counseling resources.
Stage 2: Orchestration Guardrail	LangGraph Execution (During Agent Dispatch)	• Authoritative Context Injection: Injects verified student marks and subjects from PostgreSQL. • Tool Scope Verification: Constrains A2A agent invocation to registered microservices. • Multi-Turn Memory: Preserves user facts across sessions without cross-tenant context leaks.
Stage 3: Output Guardrail	Post-Generation (Prior to Client Streaming)	• Factual Grounding: Validates that cited course codes, marks, and attendance match DB records. • Tone & Empathy Verification: Rejects punitive language; enforces constructive Socratic guidance. • Zero Risk Exposure: Strips any accidental leaks of internal risk scores or danger labels.

8. Database Schema & Data Integrity

EduGuardian maintains two isolated PostgreSQL databases: `eduguardian` (Academic ERP & Roster) and `eduguardian_chatbot` (Conversational State & Memory).

Primary Relational Entities:

- `users & teachers`: Faculty directory, departments, designations, and mentor capacities.
- `students`: Student records, USN, department, semester, section, and data source tracking.
- `mentor_assignments`: Tracks active faculty-student mentorship pairings with mentor capacity enforcement.
- `assignments & assignment_submissions`: Coursework deadlines, max marks, and student submission status.

- `attendance_records` & `quiz_results`: Subject-wise attendance % and continuous internal evaluation scores.
- `risk_predictions`: Historical audit log of calculated risk levels, recovery probabilities, and attribution factors.
- `student_resources`: Targeted PDFs and learning guides dispatched by teachers directly to students.
- `conversations & messages` (Chatbot DB): Multi-turn dialog histories with structured data and agent usage logs.

9. Key Faculty Workflows & Mentoring Pipeline

The Teacher & Admin Portal provides educators with an actionable command center to identify struggling students, manage mentoring workloads, and dispatch targeted resources.

- **Hybrid Live Roster & Multi-Tier Filtering:** Faculty can filter students across High Risk, Medium Risk, and Low Risk tabs with live search by USN, name, or semester.
- **Deep Student Risk Dossier:** Clicking any student opens their complete academic dossier—showing their 6-semester SGPA curve, attendance history, cleared supplementary backlogs, and SHAP risk factor breakdown.
- **Strict Mentor Capacity Enforcement:** Teachers have designated student capacities (e.g. 10 or 15 mentees). The system prevents overloading faculty members by dynamically tracking active mentee counts.
- **Targeted Resource Dispatch:** Teachers can upload study materials and assign them either to the entire class or targeted specifically at High-Risk students.

10. Production Deployment & Verification

EduGuardian 2.0 is packaged for instant zero-configuration deployment on any laptop or server running Docker Desktop.

Single-Command Full Stack Startup:
git clone https://github.com/nnm24is127-droid/eduguardian.git
cd eduguardian
docker compose up --build

Default Evaluation Accounts:

Role	Portal URL	Login Email / Mobile	Default Password
Class Advisor / Teacher	http://localhost:3002	preethi.salian@nitte.edu.in	123456
Subject Faculty (DCN)	http://localhost:3002	ravi.b@nitte.edu.in	123456
Student (Portal Auth)	http://localhost:3001	nnm24is127@eduguardian.ai	123456
AI Chatbot Standalone	http://localhost:3000	Auto-connects via Gateway	Shared JWT Auth

Conclusion: EduGuardian AI 2.0 provides an end-to-end, ethically grounded, and algorithmically robust platform that turns institutional examination data into early warning safeguards and actionable student recovery pathways.